



GOVT. MODEL ENGINEERING COLLEGE, THRIKKAKARA

"Evolve into academy of excellence to serve the knowledge society"

(Managed by IHRD, A Govt. of Kerala Undertaking)

DEPARTMENT OF COMPUTER ENGINEERING

**B.TECH. DEGREE SEVENTH SEMESTER
COMPUTER SCIENCE AND ENGINEERING
FIRST INTERNAL EXAMINATION – AUGUST 2025**

**Academic Year:
2025-26**

Course Code: CST401

Course Title: ARTIFICIAL INTELLIGENCE

Duration: 1½ Hr.

Max. Marks: 50

Faculty Handling the Course: **Dr. Sindhu L, Lekshmi
Subha M S, Veena Briji Philip, Sinu T S**

Course Outcomes: At the end of the course the students will be able to

CST401.1

Explain the fundamental concepts of intelligent systems and their architecture.

CST401.2

Illustrate uninformed and informed search techniques for problem solving in intelligent systems.

PART A (Answer All Questions)

Cognitive
level

CO

Marks

1 Define PEAS representation. Give the PEAS representation of a self-driving car.

L1
L2

CST401.1

3

② Distinguish between episodic and sequential environment

L2

CST401.1

3

3 Describe the state space representation of the 8-queens problem.

L1,
L2

CST401.2

3

4 Explain the significance of the branching factor in searching.

L2

CST401.2

3

5 Discuss the infrastructure needed to solve a search problem.

L1

CST401.2

3

PART B

(Answer 6 OR 7, 8 OR 9 and 10 OR 11)

6 Explain learning agent with a neat diagram.

L1,L2

CST401.1

7

OR

X Design a rational vacuum cleaner agent. Describe a rational agent function for the case in which each movement costs one point.

L1,L2

CST401.1

7

8 a) Explain the structure of a model based reflex agent with a neat diagram.

L1,L2

CST401.1

7

b) Describe the different task environment types with examples.

OR

X a) Explain the structure of a goal based reflex agent with a neat diagram.
b) Describe any four applications of AI in detail.

L1,L2

CST401.1

7

X a) Explain the working of the Uniform Cost Search algorithm with an example.

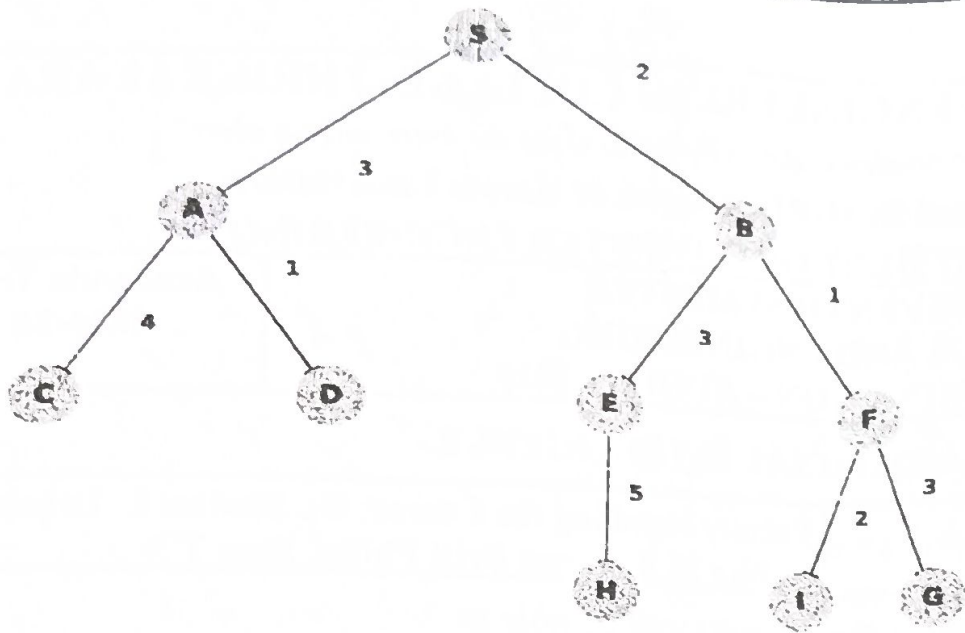
L1,L2
L3

CST401.2

7

b) Explain Best First Search algorithm. Perform BFS search in the problem given below to reach goal node G from source node S.

7



The heuristic values of S, A, B, C, D, E, F, H, I, G are 13, 12, 4, 7, 3, 8, 2, 4, 9 and 0 respectively.

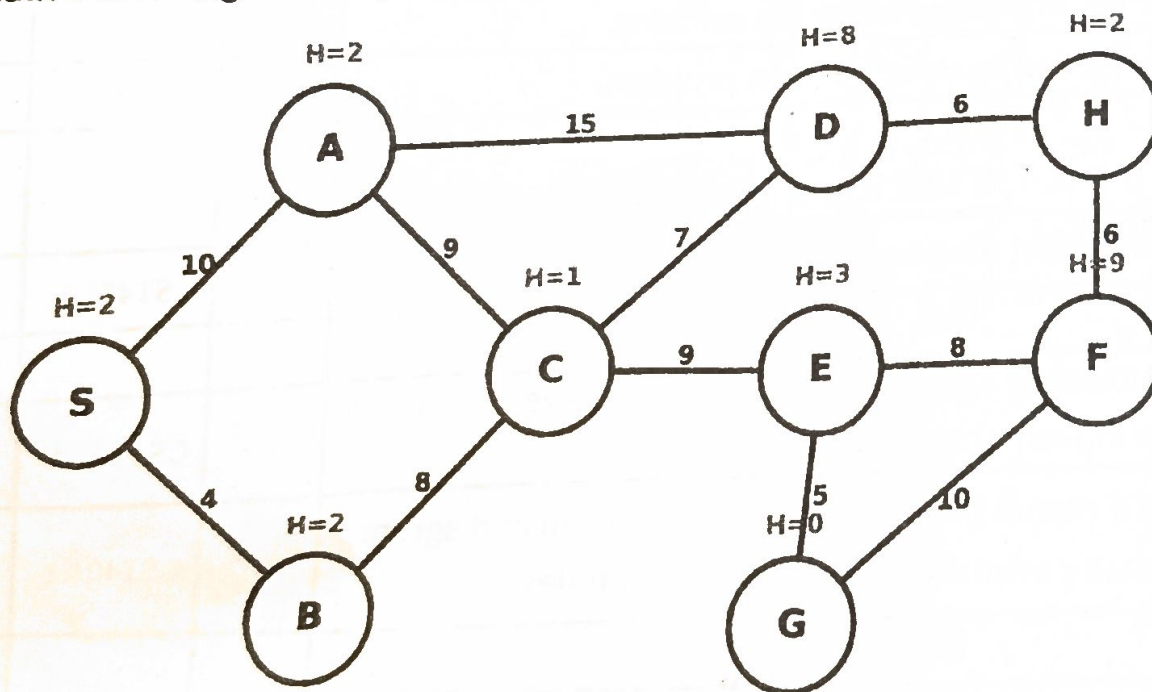
OR

a) Explain how the Iterative Deepening Search algorithm differs from Depth Limited Search algorithm. Compare and analyse the performance of both algorithms.

L1,
L2,L3

b) For the following figure S is the start state and G is the goal state. Show the order in which nodes are explored, solution path and path cost for A* algorithm.

Explain the A* algorithm. [in case of tie use lexicographic order]



CST401.2

7
7

L1-Remember

L2- Understand

L3- Apply

L4- Analyze

L5-Evaluate

L6-Create